

Algebraic Geometry – Exercise Sheet 3

Week 3: Schemes, morphisms, gluing and first global properties
Monday and Thursday afternoon sessions; Lectures 11–15

Difficulty. ★ routine; ★★ synthesis; ★★★ challenge or extension.

Use. The sheet is deliberately longer than the exercise sessions. Cerulean exercises form the suggested route; green exercises are alternatives.

For a ring R , write $\gamma_R : \mathbb{Z} \rightarrow R$, $n \mapsto n1_R$, for the unique unital ring homomorphism.

Monday afternoon — material available: Lectures 1–11

Exercise 1. Global functions as morphisms to the affine line

★ **CORE**

Let X be a scheme.

- (i) Choose an affine open covering $X = \bigcup_i U_i$. Explain why a global section $s \in \Gamma(\mathcal{O}_X)$ determines compatible morphisms $U_i \rightarrow \mathbb{A}_{\mathbb{Z}}^1$ and hence a morphism

$$f_s : X \longrightarrow \mathbb{A}_{\mathbb{Z}}^1.$$

- (ii) Show that

$$\mathrm{Hom}(X, \mathbb{A}_{\mathbb{Z}}^1) \simeq \Gamma(\mathcal{O}_X).$$

- (iii) If $X = \mathrm{Spec}(A)$, describe the morphism associated with $a \in A$.

Exercise 2. The reduced point and the double point

★ **CORE**

Inside $\mathbb{A}_{\mathbb{k}}^1 = \mathrm{Spec}(\mathbb{k}[t])$, consider

$$P = \mathrm{Spec}(\mathbb{k}[t]/(t)), \quad D = \mathrm{Spec}(\mathbb{k}[t]/(t^2)).$$

- (i) Show that the two closed immersions have the same image as subsets of $\mathbb{A}_{\mathbb{k}}^1$.
(ii) Show that P and D are not isomorphic as schemes.
(iii) Compute the maps on local rings at their unique points.

Exercise 3. A morphism between affine schemes

★ **CORE**

Let

$$\varphi : \mathbb{k}[u, v] \longrightarrow \mathbb{k}[t], \quad u \longmapsto t, \quad v \longmapsto t^2.$$

- (i) Describe the induced morphism $f : \mathbb{A}_{\mathbb{k}}^1 \rightarrow \mathbb{A}_{\mathbb{k}}^2$.
(ii) Determine $\ker(\varphi)$ and factor f through a closed subscheme of $\mathbb{A}_{\mathbb{k}}^2$.
(iii) Show that the resulting morphism onto that closed subscheme is an isomorphism.

Exercise 4. Gluing ideal sheaves and closed subschemes

★★ **CORE**

Let $X = \bigcup_i U_i$ be an open covering. Suppose that each U_i carries a quasi-coherent ideal sheaf $\mathcal{I}_i \subset \mathcal{O}_{U_i}$ and that

$$\mathcal{I}_i|_{U_i \cap U_j} = \mathcal{I}_j|_{U_i \cap U_j}$$

inside $\mathcal{O}_X|_{U_i \cap U_j}$.

- (i) Show that the $\mathcal{I}_i$ glue to an ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$.
(ii) Prove that $\mathcal{I}$ is quasi-coherent.
(iii) If $Z_i \hookrightarrow U_i$ is the closed subscheme defined by $\mathcal{I}_i$, show that the Z_i glue to the closed subscheme of X defined by $\mathcal{I}$.
(iv) Explain why none of these assertions requires X to be affine.

Exercise 5. Open subschemes and the punctured plane

★★ **CORE**

Let X be a scheme and let $U \subset X$ be open.

- (i) Prove directly from the locally affine definition that $(U, \mathcal{O}_X|_U)$ is a scheme.

(ii) Show that for every $x \in U$ one has

$$\mathcal{O}_{U,x} \simeq \mathcal{O}_{X,x}, \quad \kappa_U(x) \simeq \kappa_X(x).$$

(iii) Cover the punctured affine plane $P = \mathbb{A}_{\mathbb{k}}^2 \setminus \mathbb{W}(x, y)$ by two affine open subschemes.

(iv) Compute $\Gamma(\mathcal{O}_P)$ and prove that P is not affine.

Thursday afternoon — material available: Lectures 1–14

Exercise 6. Dual numbers and tangent vectors

★★ CORE

Let $D = \text{Spec}(\mathbb{k}[\varepsilon]/(\varepsilon^2))$ and let $p = (a, b) \in \mathbb{A}_{\mathbb{k}}^2(\mathbb{k})$.

(i) Show that morphisms $D \rightarrow \mathbb{A}_{\mathbb{k}}^2$ whose closed point maps to p are uniquely of the form

$$x \mapsto a + u\varepsilon, \quad y \mapsto b + v\varepsilon.$$

(ii) Let $C = \mathbb{W}(f) \subset \mathbb{A}_{\mathbb{k}}^2$. Show that such a morphism factors through C precisely when

$$f(a, b) = 0, \quad f_x(a, b)u + f_y(a, b)v = 0.$$

(iii) Apply this to the parabola $y = x^2$.

(iv) Assume $\text{char}(\mathbb{k}) \neq 2, 3$. For the cusp $C = \mathbb{W}(y^2 - x^3)$, compare the tangent spaces at $(0, 0)$ and $(1, 1)$. Which point is singular and which is smooth according to the Jacobian criterion?

Exercise 7. The line with two origins

★★ CORE

Glue two copies

$$X_1 = \text{Spec}(\mathbb{k}[t_1]), \quad X_2 = \text{Spec}(\mathbb{k}[t_2])$$

along $D(t_1) \simeq D(t_2)$ by $t_1 \mapsto t_2$. Let X be the resulting scheme.

(i) Describe the underlying topological space.

(ii) Show that $\Gamma(\mathcal{O}_X) \simeq \mathbb{k}[t]$.

(iii) Show that the canonical map $X \rightarrow \text{Spec } \Gamma(\mathcal{O}_X)$ identifies the two origins.

(iv) Deduce that X is not affine.

Exercise 8. Rational and closed points of the projective line

★★ CORE

Construct $\mathbb{P}_{\mathbb{k}}^1$ by gluing $U_0 = \text{Spec}(\mathbb{k}[t])$ and $U_1 = \text{Spec}(\mathbb{k}[s])$ along $s = t^{-1}$.

(i) For every field extension $K/\mathbb{k}$, prove that

$$\mathbb{P}_{\mathbb{k}}^1(K) = \mathbb{P}_K^1.$$

Describe the points in the two standard charts.

(ii) Describe all closed points of $\mathbb{P}_{\mathbb{k}}^1$, compute their residue fields, and determine which are $\mathbb{k}$ -rational.

(iii) For every closed point x , exhibit its canonical $\kappa(x)$ -valued point and verify that $\kappa(x)/\mathbb{k}$ is finite. Give a field-valued point whose image is not closed.

(iv) Prove that $\Gamma(\mathcal{O}_{\mathbb{P}_{\mathbb{k}}^1}) = \mathbb{k}$.

(v) Deduce that $\mathbb{P}_{\mathbb{k}}^1$ is not affine.

Exercise 9. The projective conic from its standard opens

★★ CORE

Let

$$S = \mathbb{k}[x_0, x_1, x_2]/(x_0x_2 - x_1^2), \quad C = \text{Proj}(S).$$

(i) Compute $(S[x_0^{-1}])_0$ and $(S[x_2^{-1}])_0$.

(ii) Identify $D_+(x_0)$ and $D_+(x_2)$ with affine lines.

(iii) Describe their overlap and recover the gluing rule for $\mathbb{P}^1$.

- (iv) Show that projection from $p = [1 : 0 : 0]$ extends to a morphism $\pi : C \rightarrow \mathbb{P}_{\mathbb{k}}^1$ by using the formulas $[x_0 : x_1]$ on $D_+(x_0)$ and $[x_1 : x_2]$ on $D_+(x_2)$. Prove that its inverse is $[s : t] \mapsto [s^2 : st : t^2]$.

Exercise 10. Base loci: projection and Cremona

*** CHOICE

- (i) Consider

$$\pi : \mathbb{P}_{\mathbb{k}}^2 \dashrightarrow \mathbb{P}_{\mathbb{k}}^1, \quad [x_0 : x_1 : x_2] \longmapsto [x_1 : x_2].$$

Determine its domain of definition and its base locus.

- (ii) In $\mathbb{P}_{\mathbb{k}}^2 \times \mathbb{P}_{\mathbb{k}}^1$, show that the closure of its graph is defined by $x_1v - x_2u = 0$. Describe the fibre of the first projection over $[1 : 0 : 0]$.
- (iii) For the standard Cremona transformation

$$[x_0 : x_1 : x_2] \dashrightarrow [x_1x_2 : x_0x_2 : x_0x_1],$$

compute the base locus and show that the transformation is its own inverse on the dense torus $x_0x_1x_2 \neq 0$.

Exercise 11. Connectedness beyond affine schemes

** CORE

Let X be a scheme.

- (i) Prove that X is disconnected if and only if $\Gamma(\mathcal{O}_X)$ has a non-trivial idempotent.
- (ii) Use the global-function computations from this sheet to show that $\mathbb{P}_{\mathbb{k}}^1$ and the line with two origins are connected.
- (iii) Compare the connected reducible scheme $\text{Spec}(\mathbb{k}[x, y]/(xy))$ with the disconnected scheme $\text{Spec}(\mathbb{k} \times \mathbb{k})$.

Exercise 12. A doubled projective line and its reduction

*** CORE

Let

$$U_0 = \text{Spec}(\mathbb{k}[t, \varepsilon]/(\varepsilon^2)), \quad U_1 = \text{Spec}(\mathbb{k}[s, \eta]/(\eta^2)).$$

Glue $D(t) \subset U_0$ to $D(s) \subset U_1$ using $s = t^{-1}$ and $\eta = \varepsilon$, and call the resulting scheme X .

- (i) Show that $(\varepsilon) \subset \mathcal{O}_{U_0}$ and $(\eta) \subset \mathcal{O}_{U_1}$ glue to a quasi-coherent ideal sheaf $\mathcal{I} \subset \mathcal{O}_X$ with $\mathcal{I}^2 = 0$.
- (ii) Describe X_{red} and prove that $X_{\text{red}} \simeq \mathbb{P}_{\mathbb{k}}^1$.
- (iii) Show that every morphism from a reduced scheme Y to X factors uniquely through $X_{\text{red}} \hookrightarrow X$.
- (iv) Deduce that X is not affine.

Exercise 13. A weighted projective chart

*** OPTIONAL

Let $R = \mathbb{k}[x, y, z]$ with $\deg x = \deg y = 1$ and $\deg z = 2$, and set $X = \text{Proj}(R)$. Show that $D_+(x)$ and $D_+(y)$ are affine planes and that

$$D_+(z) \simeq \text{Spec}(\mathbb{k}[a, b, c]/(ac - b^2)).$$

Identify the singular point of this chart.

After Friday's lecture or preparation for Week 4

Exercise 14. Components and generic points

** CHOICE

Let

$$I = (xy, xz) \subset \mathbb{k}[x, y, z], \quad X = \text{Spec}(\mathbb{k}[x, y, z]/I).$$

Show that $I = (x) \cap (y, z)$. Determine the irreducible components, their generic points and dimensions, and decide whether X is connected or integral.

Exercise 15. The function field of the cusp

** CHOICE

Let

$$C = \text{Spec}(\mathbb{k}[x, y]/(y^2 - x^3)).$$

Show that C is integral and that $K(C) \simeq \mathbb{k}(t)$.